Note that the practice final exam has two parts: (1) multiple-choice/numerical questions and (2) problems. This file only contains the solutions to the problems. The solutions for multiple-choice questions can be found on Canvas.

Part I: Multiple-choice/numerical questions: Please see Canvas.

Part II: Problems:

1. Masson Corporation is evaluating the opportunity to acquire 100% Chancery Inc.'s equity. You are given the following information:

	Masson Corporation (Acquirer)	Chancery Inc. (Target)
Shares outstanding	20 million	50 million
Standalone share value before the M&A (same as the pre-M&A market value per share)	\$25 per share	\$5 per share
Total market value of outstanding bonds before the M&A	\$100 million	\$100 million

Both the acquirer and the target have no other liabilities other than the bonds listed above. The acquirer can issue new shares at \$24 per share in stock consideration.

The management of Masson Corporation further estimates that the combined total firm value including merger gains will be \$1,050 million.

A) What is the NPV of the M&A deal from the acquirer's perspective if the final offer price is \$8 per share paid with full cash?

B) What should the maximum total offer price acceptable to the acquirer be?

C)If the acquirer decides to pay the maximum offer price with 100% stock consideration, what should be the exchange ratio?

A)

Step 1: Calculate the merger gains, the total NPV generated by the M&A deal

- Acquirer standalone total firm value: V_A= D+E = 100+20m*25=\$600 million
- Target standalone total firm value: = V_B= D+E = 100+50m*5 = \$350 million
- Merger gains = V_{AB} – (V_A + V_B) = 1050 million – (600 million + 350 million) = \$100 million
- Note that merger gains are the total NPV generated by the M&A deal and will be shared between the acquirer and the target.

Step2: Target's piece of merger gains

- The target's piece of merger gains or the target's piece of NPV
- = total acquisition premium = total offer price – the target's market value of equity
- =shares outstanding x offer price per share – shares outstanding x share value before the M&A
- = 8 x 50 – 5 x 50 = \$150 million.

Step 3: Acquirer's piece of merger gains

- The acquirer's piece of NPV = merger gains – the target's piece of NPV
- = $100 million - $150 million = - $50.
- In other words, the M&A destroys the acquirer's shareholder value because the offer price is too high.

B)

The maximum offer price is the offer price at which the target gets the entire merger gains and the acquirer gets nothing.

- Therefore, the maximum total offer price = the target's standalone equity + merger gains = $5 x 50 + $100 =$350 million.

C)

- Offer price per share = $350/50 = $7 per share
- Exchange ratio = how many acquirer's shares to be paid for each share of the target
- = offer price in stock consideration per share/the acquirer's issuance price = $7/24 = 0.29

2. You are provided with Nexus Innovations's historical income statement and balance sheet for the year 0 (assume today is the year 0 end),, as well as its Pro-forma financial statements for the year 1:

Income Statement (year 0 and year 0, in thousand dollars)		
	Actual	**Pro Forma**
Year	**Year 0**	**Year 1**
Sales	$97,194	$113,557
Cost of Goods Sold		
Direct Material	($20,532)	($23,752)
Direct Labour	($23,784)	($28,333)
Gross Profit	$52,878	$61,472
Sales and Marketing	($16,037)	($20,440)
Administrative	($14,579)	($17,034)
EBITDA	$22,262	$23,999
Depreciation	($5,995)	($5,946)
EBIT	$16,267	$18,053
Net Interest Expenses	($6,600)	($6,600)
EBT	$9,667	$11,453
Income Tax (30%)	($2,900)	($3,436)
Net Income	$6,767	$8,017

Balance Sheet (year 0 and year 1, in thousand dollars)		
	Actual	**Pro Forma**
Year	**Year 0**	**Year 1**
Assets		
Operating Cash	$15,000	$15,000
Accounts Receivable	$8,966	$13,001
Inventories	$7,151	$8,374
Total Current Assets	$31,117	$36,375
Property, Plant, & Equipment	$139,565	$136,565
Total Assets	$170,682	$172,940
Liabilities		
Account Payable	$6,085	$7,313
Long-term Debt	$110,000	$110,000
Total Liabilities	$116,085	$117,313
Shareholder's Equity		

Total Shareholders' Equity	$54,597	$55,627
Total Liabilities and Equity	$170,682	$172,940

The expected capital expenditure in year 1 is $3946 (in thousand dollars). Nexus Innovations has a constant market-based D/E ratio of 0.5, a corporate tax rate of 30%, and a pre-tax cost of debt of 5%. The risk-free rate is 3%. The expected market risk premium is 6%. Nexus Innovations's unlevered beta is 0.5.

A) What is the firm's WACC?

B) What is the firm's FCFF in year 1?

C) What is the standalone value of the firm's standalone equity value as of the year 0 end? It is expected that the FCFF will grow at a constant rate of 2.5% from year 1. Also, assume that the company does not have any non-operating assets and that the market value of the company's debt is the same as its book value.

A) Calculate the WACC of the company

- Step 1: cost of equity
  - $\beta_L = \beta_U \times \left(1 + \frac{D}{E} \times (1 - T)\right)$
  - $\beta_L = 0.5 \times \left(1 + 0.5 \times (1 - 0.3)\right) = 0.675$
  - $r_e = r_f$+beta*Market Risk Premium = 3%+0.675*6%=7.05%
- Step 2: calculate the weights of debt and equity
  - If D/E=0.5, then
    - W_D = D/(D+E)=0.5/(0.5+1) =1/3
    - W_E = E/(D+E)=1/(0.5+1)=2/3
- Step 3: calculate the WACC
  - WACC =W_D*R_D*(1-T_C)+W_E*r_e=1/3*5%*0.7+2/3*7.05%=5.87%

B) Calculate the expected FCFF in year 1

Note we should exclude non-operating cash in net working capital calculation (in this question, we are only provided with operating cash, so we don't need to do anything with it).

Step 1: Change in NWC

- NWC in Year 1 = (Operating Cash + A.R. + Inventories) – A.P. = 15,000 + 13,001+8,374 - 7,313 = 29,062.
- NWC in Year 0 = (Operating Cash + A.R. + Inventories) – A.P. = 15,000 + 8,966 +7,151 - 6,085 = 25,032.
- Change in NWC = NWC of year 1 – NWC of year 0 = 29,062 - 25,032 = $4,030.

Step 2: FCFF

- FCFF = EBIT*(1-Tc) + Depreciation & Amortisation – ΔNWC – Capital Expenditure
- FCFF = 18,053*(1-0.3) + 5946 – 4030 -3946 = 10,607

Step 3: Estimate the total standalone value of the firm's operating assets (note that there could be some rounding errors in the calculation):

- In fact, all the future cash flows will be growing at a constant rate g. Therefore the total standalone value of the firm's operating should be the PV of a growing perpetuity:
- V (operating assets) = 10,607/(5.87%- 2.5%)=$315,062 (Thousands)
- Note that there could be some rounding error (which is fine) in your final answer.

Step 4: Estimate the total standalone equity value

- Total standalone firm value = value of operating assets (from step 3) + value of non-operating assets = $315,062 (Thousands)+0 = $315,062 (Thousands).
- Equity Value = Total firm value – Mkt value of debt = $315,062 -110,000=$205,062.38 (thousands)

3. Your company plans to acquire 100% of Harrison & Sons Manufacturing's equity.

You are given the following forecasts of the target firm when merger gains are excluded:

- The FCFF one year from today ($FCFF_1$) will be $5 million.
- The FCFF will grow at a constant rate of 3% per year perpetually.
- The target's unlevered beta is 0.6.
- The target's cost of debt is 5%.
- The target will main a D/E ratio of 0.5 if not acquired.

You are also given the following forecasts of the target firm when merger gains are included:

- The FCFF one year from today ($FCFF_1$) will be $5 million.
- The FCFF will grow at a constant rate of 3.5% per year perpetually due to some soft synergy.
- Due to some financial synergy, the acquirer's cost of debt is 4% and will maintain a D/E of 1 after the M&A.

The risk-free rate is 3%. The market risk premium is 6%. The corporate tax rate is 30%. The target firm has zero non-controlling interests and zero non-operating assets. What is the total value of all these merger gains?

Step 1: The Standalone value of the target

- <u>Step 1.1: WACC</u>
    - Target's standalone firm value means the target's status quo firm value. Therefore, we should use the target's inputs.
    - Target's levered beta $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right] = 0.6 \times [1 + 0.5 \times (1 - 0.3)] = 0.81$
    - Target's cost of equity $r_e = r_f + \beta_e \times MRP = 3\% + 0.81 \times 6\% = 7.86\%$
    - Target's WACC
        - $W_d = \frac{D}{D+E} = \frac{0.5}{0.5+1} = 1/3$

- $W_e = \frac{E}{D+E} = \frac{1}{0.5+1} = 2/3$
- $WACC = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 1/3 \times 5\% \times (1 - 0.3) + 2/3 \times 7.86\% = 6.41\%$

- Step 1.2: Target standalone value
  - The FCFFs will grow at a constant rate, so they are a perpetuity.
  - $V(\text{operating assets}) = \frac{FCFF_1}{r-g} = \frac{5}{0.064-0.03} = \$146.771\ \text{mln}$

Step 2: The value of the target including synergies

- Step 2.1: WACC
  - There is no change in the target's business nature, so we should use the target's unlevered beta.
  - Target's levered beta using **the acquirer's D/E** : $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right] = 0.6 \times [1 + 1 \times (1 - 0.3)] = 1.02$
  - Target's cost of equity $r_e = r_f + \beta_e \times MRP = 3\% + 1.02 \times 6\% = 9.12\%$
  - Target's WACC using **the acquirer's r_d and D/E**
    - $W_d = \frac{D}{D+E} = \frac{1}{1+1} = 0.5$
    - $W_e = \frac{E}{D+E} = \frac{1}{1+1} = 0.5$
    - $WACC = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0.5 \times 4\% \times (1 - 0.3) + 0.5 \times 9.12\% = 5.96\%$
- Step 2.2: Target's firm value, including the synergies
  - Due to the financial synergy, the WACC will be 5.96%
  - Due to the soft synergy, g=5%
  - $V(\text{operating assets}) *= \frac{FCFF_1}{r-g} = \frac{5}{0.0596-0.035} = \$203.252\ \text{mln}$

Step 3: The value of the these merger gains

- $Total\ Synergy\ Value = V(operating\ assets) * - V(operating\ assets) = \$203.252 - \$146.771 = \$56.48\ mln$

4. You are provided the following information with regard to a private target company's standalone value:

- The free cash flow one year from today (FCFF1) will be $6 million.
- The FCFF will grow at a constant rate of 3% per year perpetually.
- The target company is currently unlevered.

You find three listed companies that have a similar business nature to the company that you are valuing. You run regressions to estimate the equity beta of the three comparable companies and get the following information:

Comparable companies	Equity Beta	D/E
Integral Dynamics Corporation	1.08	0.5
Spectrum Global Enterprises	1.092	0.8
Foundation Capital Partners	1.02	1.0

The risk-free rate is 3%. The market risk premium is 6%. The corporate tax rate is 30%. The target firm has zero non-controlling interests and zero non-operating assets. Once the M&A is completed, the acquirer will keep a D/E ratio of 1 and have a cost of debt of 5%. The M&A deal has no impact on the FCFFs and the perceptual growth rate. What is the NPV of the M&A deal from the acquirer's perspective if the acquirer pays a total acquisition premium of $10 million?

Step 1: Target's standalone total firm value

- 1.1 Calculate the average unlevered beta of the three comparables:
  - X: $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
    - $1.08 = \beta_u \times [1 + 0.5 \times (1 - 0.3)]$
    - $\beta_u = 0.8$
  - Y: $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
    - $1.092 = \beta_u \times [1 + 0.8 \times (1 - 0.3)]$
    - $\beta_u = 0.7$
  - Z: $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
    - $1.02 = \beta_u \times [1 + 1 \times (1 - 0.3)]$
    - $\beta_u = 0.6$
  - Average unlevered beta= (0.8+0.7+0.6)/3=0.7

- 1.2: WACC of the target for standalone valuation
  - We assume that the target's unlevered beta is the same as the average unlevered beta of the three comparables. $\beta_u = 0.7$
  - Since the target is unlevered, the target's equity beta $\beta_e = \beta_u = 0.7$
  - Target's cost of equity $r_e = r_f + \beta_e \times MRP = 3\% + 0.7 \times 6\% = 7.2\%$
  - Target's WACC
    - $W_d = \frac{D}{D+E} = 0$
    - $W_e = \frac{E}{D+E} = \frac{1}{0+1} = 1$
    - $WACC = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0 + 7.2\% = 7.2\%$

- 1.3: Target standalone total firm value

- $\text{The FCFFs will grow at a constant rate, so they are a perpetuity.}$
- $\text{V(total assets)} = \frac{\text{FCFF}_1}{r-g} = \frac{6}{0.072-0.03} = \$142.856\text{mln}$

<u>Step 2: Target's total firm value including all merger gains</u>

- <u>2.1 Calculate the re-levered beta and cost of equity of the target using the acquirer's D/E:</u>
  - $\beta_L = \beta_u \times \left[1 + \frac{D}{E} \times (1 - T_C)\right]$
  - $\beta_L = 0.7 \times [1 + 1 \times (1 - 0.3)]$
  - $\beta_L = 1.19$
  - $\text{Target's cost of equity } r_e = r_f + \beta_e \times \text{MRP} = 3\% + 1.19 \times 6\% = 10.14\%$

- <u>2.2: WACC of the target using the acquirer's D/E and cost of debt</u>
  - WACC
    - $W_d = \frac{D}{D+E} = \frac{1}{1+1} = 0.5$
    - $W_e = \frac{E}{D+E} = \frac{1}{1+1} = 0.5$
    - $\text{WACC} = W_d \times r_d \times (1 - T_C) + W_e \times r_e = 0.5 \times 5\% \times (1 - 0.3) + 0.5 \times 10.14\% = 6.82\%$
- <u>2.3: Target's total firm value including all merger gains</u>
  - $\text{The FCFFs will grow at a constant rate, so they are a perpetuity.}$
  - $\text{V(operating assets)} = \frac{\text{FCFF}_1}{r-g} = \frac{6}{0.0682-0.03} = \$157.0681\text{mln}$

<u>Step 3: Valuation of merger gains</u>

- Target's firm value including merger gains minus target's standalone firm value
- = 157.0181 – 142.8571 = $14.21 million.
- Note this amount is the total NPV of the M&A deal.

<u>Step 4: NPV of the M&A from the acquirer's perspective:</u>

- NPV of the M&A from the target's perspective = total acquisition premium =$10 million dollar.
- NPV of the M&A from the acquirer's perspective = total NPV from step 3 minus – 10 million =14.21 – 10 = $4.21 million.